Sr.No	Title	Author	Journal	Year	Model / Methodology	Data Source	Results	ST	MT	LT	Citation	Patent
1	Large Language Model-Empowered Interactive Load Forecasting	Yu Zuo, Dalin Qin, Yi Wang	arXiv Preprint	2025	LLM-based multi-agent forecasting pipeline integrating human feedback	Two real-world load datasets	Interactive forecasting improves accuracy by integrating operator insights into ML pipeline	Yes	No	No	N/A	No
2	Short-Term Power Demand Forecasting for Diverse Consumer Types to Enhance Grid Planning and Synchronisation	Asier Diaz-Iglesias et al.	arXiv Preprint	2025	ML models with clustering for consumer-type classification (industrial, residential, commercial)	Copernicus ERA5 weather data + smart meter data	Customized models for each consumer group significantly improve STLF accuracy	Yes	No	No	N/A	No
3	Graph Neural Networks for Electricity Load Forecasting	Eloi Campagne et al.	arXiv / EDF Research	2025	Graph Neural Networks (GCN, GraphSAGE, GAT) with attention and ensemble aggregation	Synthetic + France regional + UK datasets	Graph-aware models outperform conventional neural baselines and capture spatial dependencies	Yes	Yes	No	N/A	No
4	Load Forecasting on a Highly Sparse Electrical Load Dataset Using Gaussian Interpolation	Chinmoy Biswas et al.	Electric Power Systems Research (preprint)	2025	Gaussian interpolation + LSTM / BiLSTM / Transformer comparison	BUET campus load dataset + PJM dataset	LSTM performs best after interpolation on highly sparse datasets (62% missing data)	Yes	No	No	N/A	No
5	DemandCast: Global Hourly Electricity Demand Forecasting	Kevin Steijn et al.	NeurIPS Climate ML Workshop	2025	Gradient Boosting (XGBoost) forecasting framework	Global electricity demand + ERA5 temperature + socioeconomic data	Accurate global demand prediction framework scalable across countries	Yes	Yes	Yes	N/A	No
6	A Lightweight Multi-View Approach to Short-Term Load Forecasting	Julien Guité-Vinet et al.	Elsevier Preprint	2026	Multi-view embedding-based deep learning model with feature dropout	Multiple real-world time series datasets	Comparable performance to large transformer models with fewer parameters	Yes	No	No	N/A	No
7	Assessing Electricity Demand Forecasting with Exogenous Data in Time Series Foundation Models	Wei Soon Cheong et al.	AAAI Preprint	2026	Time Series Foundation Models (Chronos, MOIRAI, TTM) vs LSTM baseline	Singapore and Australian electricity markets	LSTM baseline sometimes outperforms foundation models in stable climates	Yes	Yes	No	N/A	No
8	Bayesian Transformer for Probabilistic Load Forecasting in Smart Grids	Sajib Debnath et al.	arXiv Preprint	2026	Bayesian Transformer with MC Dropout + Variational inference	PJM, ERCOT, ENTSO-E datasets + NOAA weather	Achieves CRPS improvement of 7.4% over deep ensembles with calibrated uncertainty intervals	Yes	Yes	No	N/A	No
9	Reliable Grid Forecasting: State Space Models for Safety-Critical Energy Systems	Sunki Hong, Jisoo Lee	IEEE Preprint	2026	State Space Models (Mamba-based architectures) for reliable forecasting	CAISO dataset with weather integration	Improved reliability metrics and reduced tail risk in grid forecasting	Yes	Yes	No	N/A	No
10	Resilient Load Forecasting under Climate Change: Adaptive Conditional Neural Processes	Chenxi Hu et al.	arXiv Preprint	2026	Adaptive Conditional Neural Process (AdaCNP) probabilistic model	Real-world power system datasets	Reduces MSE by 22% during extreme events compared to baselines	Yes	Yes	No	N/A	No
11	Short-Term Electricity Load Forecasting with Multi-Frequency Reconstruction Diffusion	Qi Dong et al.	Information Sciences Preprint	2026	Diffusion-based model with multi-frequency reconstruction and LSTM-Transformer denoising	AEMO and ISO-NE datasets	Diffusion approach improves prediction stability and noise robustness	Yes	No	No	N/A	No
12	From ARIMA to Attention: Power Load Forecasting Using Temporal Deep Learning	Suhasnadh Reddy Veluru et al.	arXiv Preprint	2026	Comparative study of ARIMA, LSTM, BiLSTM, Transformer	PJM Hourly Energy Consumption dataset	Transformer achieves best performance with MAPE 3.8%	Yes	No	No	N/A	No